

HW8: Optimal FIR design(Part 2)

Due [Wednesday 29 December 23:59](#)

Note: Because in this homework we still need to use **myChebyPol()**, please include this function in your folder as well.

Task (60%): Please write a function called **newArgument()** that automatically returns the local extrema of the polynomial $P(x)$.

```
[arg, mag] = newArgument(p_x, xplotloc, L, delta, x_p, x_s);
```

- [1] Your newArgument() must output these two vectors: **arg** and **mag**.
- arg is the argument where local extrema occur. It is in the range $[-1, 1]$.
 - mag is the magnitude of the local extrema points. Don't forget to include ω_p and ω_s in your arg and mag.
- [2] Your newArgument() have inputs: **p_x**, **xplotloc** and **delta**.
- p_x: the polynomial
 - xplotloc: x-axis of p_x
 - delta: Assume passband error equals stopband error

Task (20%): Please plot the optimal polynomial. Also, please plot the magnitude response given the impulse response $h[n]$, and mark the places of ω_p and ω_s . For both plots above, please use the following parameters:

```
omega_p = 0.2*pi;  
omega_s = 0.24*pi;  
L = 32;
```

Discussion (20%): continuing from above, discuss any difficulties you encountered in the homework.

The following are some examples.

- **newArgument()** should returns these orange points.

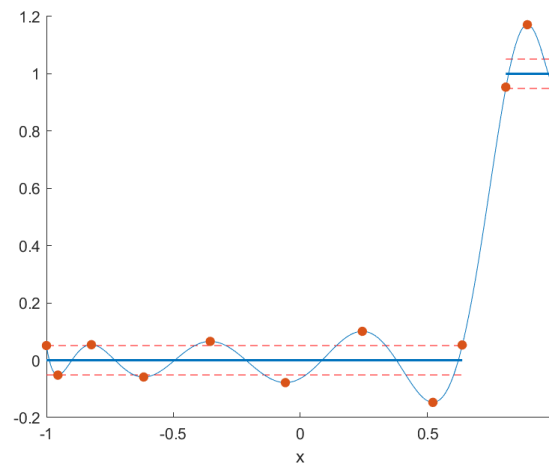


Figure 1: the polynomial before the optimization

- Below is an example of correct implementation for you to know what to anticipate. This is how the polynomial looks like when it reaches its optimum **after several iterations**.

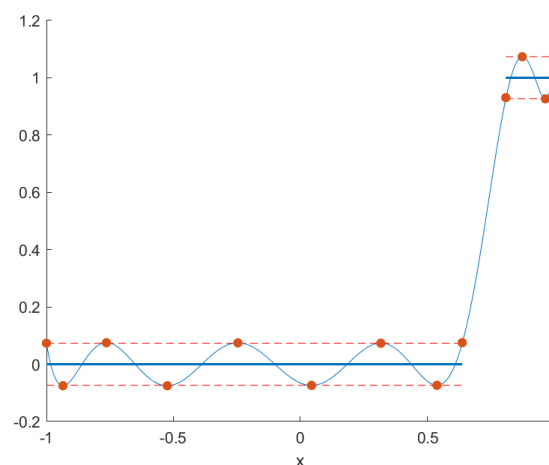


Figure 2: the polynomial after the optimization

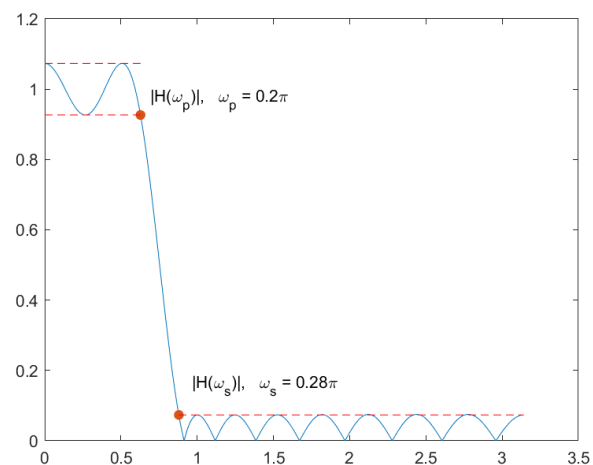


Figure 3: the magnitude response after the optimization